

Devops Lab Project 8-
Aarohi Kondpalle



The image shows a code editor window with a dark theme. At the top, there are two tabs: 'Project7_AarohiKondpalle.pdf' and 'app.py'. The breadcrumb navigation shows the path: 'Devops-Lab-L1_2023-27 > 23070122004_AarohiKondpalle > Project8 > user-service > app.py'. The code is a Python Flask application. It imports Flask and jsonify. It creates an app instance. There are two routes: a home route at '/' that returns a JSON object with 'service': 'User Service' and 'status': 'running', and a users route at '/users' that returns a JSON list of three users: Aarohi, Rahul, and Priya. The app is run on host '0.0.0.0' and port 5001.

```
1 from flask import Flask, jsonify
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     return jsonify({
8         "service": "User Service",
9         "status": "running"
10    })
11
12 @app.route("/users")
13 def users():
14     return jsonify([
15         {"id": 1, "name": "Aarohi"},
16         {"id": 2, "name": "Rahul"},
17         {"id": 3, "name": "Priya"}
18    ])
19
20 if __name__ == "__main__":
21     app.run(host="0.0.0.0", port=5001)
22
```

```
View Go Run Terminal Help devtemp
X Dockerfile U X
Devops-Lab-L1_2023-27 > 23070122004_AarohiKondpalle > Project8 > user-service > Dockerfile
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6
7 RUN pip install --no-cache-dir -r requirements.txt
8
9 COPY app.py .
10
11 EXPOSE 5001
12
13 CMD ["python", "app.py"]
```

Project8	•	🧪
frontend	•	
app.py	U	🤖
Dockerfile	U	
requirements.txt	U	
order-service	•	
app.py	U	
Dockerfile	U	
requirements.txt	U	
product-service	•	
app.py	U	
Dockerfile	U	
requirements.txt	U	
user-service	•	
app.py	U	
Dockerfile	U	
requirements.txt	U	

```
localhost:5001/users

pretty-print

{"id":1,"name":"Aarohi"}, {"id":2,"name":"Rahul"}, {"id":3,"name":"Priya"}]
```

```
localhost:5002/products

Pretty-print

[{"id":1,"name":"Laptop","price":55000}, {"id":2,"name":"Smartphone","price":25000}, {"id":3,"name":"Headphones","price":3000}]
```

```
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker build -t user-service:1.0 ./user-service
[+] Building 0.5s (1/2)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.2s
[+] Building 0.6s (1/2)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.2s
[+] Building 0.8s (1/2)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.2s
[+] Building 0.9s (1/2)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.2s
[+] Building 1.1s (1/2)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.2s
```

```
user-service:1.0                                     25206865576a      211MB      51.3MB
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker run -d --name user-service-test -p 5001:5001 user-service:1.0
c93e9c75b2b3567d8f16eac8540c5d72bf9c75e2071b4ab785d7757ff153bcd
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker stop user-service-test
>> docker rm user-service-test
user-service-test
user-service-test
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker build -t product-service:1.0 ./product-service
[+] Building 8.3s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 217B                                              0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim              2.2s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6 0.0s
```

```
retail-web-app:2.0                                     e34daa263946      549MB      188MB      U
user-service:1.0                                     25206865576a      211MB      51.3MB
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker run -d --name product-service-test -p 5002:5002 product-service:1.0
7054472b78539486ade5786eb9c2fac185cc5d9dd79c4f96f3458f9aa632e7aa
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker stop product-service-test
product-service-test
product-service-test
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8>
```

```
product-service-test
product-service-test
PS C:\Users\aaaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker build -t order-service:1.0 ./order-service
[+] Building 3.0s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 217B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6
=> => resolve docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6
=> [internal] load build context
=> => transferring context: 666B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY app.py
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:e2d774fb2ed0d89dab278a5c39bc969e2ffa99af161eea5593a10e203c7fc899
=> => exporting config sha256:33567cd736746d4e59af74c976dcb891bfabe9837547f9e697ed5da22ee934a4
=> => exporting attestation manifest sha256:5871d23cb872486a1a1a221534ea9753b7cc33ff47c81bcb2431baf496cbe25
=> => exporting manifest list sha256:dc47bff48a015c3d457551cbe7a28aa2a359602f07ec10b7bce53e7bf1526e
=> => naming to docker.io/library/order-service:1.0
=> => unpacking to docker.io/library/order-service:1.0
PS C:\Users\aaaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> docker run -d --name order-service-test -p 5003:5003 order-service:1.0
d8ba76a17f4cd704d59a78b3ad148be10e303c6e2b57cb2289005e1a6edc1112
PS C:\Users\aaaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8>
```

```
localhost:5003/orders
Pretty-print
[{"id":1,"product":"Laptop","status":"Confirmed","user":"Aarohi"}, {"id":2,"product":"Smartphone","status":"Shipped","user":"Rahul"}, {"id":3,"product":"Headphones","status":"Delivered","user":"Priya"}]
```

```
FROM python:3.11-slim
apiVersion: v1
kind: Secret
metadata:
  name: microservices-secret
type: Opaque
stringData:
  API_KEY: "devops-project8-key"
PS C:\Users\aaaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get configmaps
NAME                                DATA  AGE
kube-root-ca.crt                   1      39d
loki-loki-stack                    1      39d
loki-loki-stack-test               1      39d
microservices-config               3      15s
mongo-config                       1      41m
monitoring-grafana                 1      39d
monitoring-grafana-config-dashboards 1      39d
monitoring-kube-prometheus-alertmanager-overview 1      39d
monitoring-kube-prometheus-apiserver 1      39d
monitoring-kube-prometheus-cluster-total 1      39d
monitoring-kube-prometheus-controller-manager 1      39d
monitoring-kube-prometheus-etcd    1      39d
monitoring-kube-prometheus-grafana-datasource 1      39d
monitoring-kube-prometheus-grafana-overview 1      39d
monitoring-kube-prometheus-k8s-coredns 1      39d
monitoring-kube-prometheus-k8s-resources-cluster 1      39d
monitoring-kube-prometheus-k8s-resources-multicluster 1      39d
monitoring-kube-prometheus-k8s-resources-namespace 1      39d
```

```
http://127.0.0.1:26194/dashboard
Pretty-print
{"orders":[{"id":1,"product":"Laptop","status":"Confirmed","user":"Aarohi"}, {"id":2,"product":"Smartphone","status":"Shipped","user":"Rahul"}, {"id":3,"product":"Headphones","status":"Delivered","user":"Priya"}], "products":[{"id":1,"name":"Laptop","price":55000}, {"id":2,"name":"Smartphone","price":25000}, {"id":3,"name":"Headphones","price":3000}], "users":[{"id":1,"name":"Aarohi"}, {"id":2,"name":"Rahul"}, {"id":3,"name":"Priya"}]}
```

```

PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get services
monitoring-kube-prometheus-alertmanager ClusterIP 10.108.179.96 <none> 9093/TCP,8080/TCP 39d
monitoring-kube-prometheus-operator ClusterIP 10.106.187.61 <none> 443/TCP 39d
monitoring-kube-prometheus-prometheus ClusterIP 10.101.178.195 <none> 9090/TCP,8080/TCP 39d
monitoring-kube-state-metrics ClusterIP 10.96.169.70 <none> 8080/TCP 39d
monitoring-prometheus-node-exporter ClusterIP 10.109.46.94 <none> 9100/TCP 39d
prometheus-operated ClusterIP None <none> 9090/TCP 39d
social-media-service NodePort 10.98.118.26 <none> 80:30577/TCP 13d
user-service ClusterIP 10.111.175.128 <none> 5001/TCP 6s
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> notepad product-service.yaml
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl apply -f product-service.yaml
service/product-service created
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get service product-service
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
product-service ClusterIP 10.98.159.227 <none> 5002/TCP 6s
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> notepad order-service.yaml
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl apply -f order-service.yaml
service/order-service created
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get service order-service
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
order-service ClusterIP 10.106.103.190 <none> 5003/TCP 7s
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> notepad frontend-service.yaml
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl apply -f frontend-service.yaml
service/frontend-service created
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
alertmanager-operated ClusterIP None <none> 9093/TCP,9094/TCP,9094/UDP 39d
frontend-service NodePort 10.104.92.39 <none> 5000:30001/TCP 6s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 39d
loki ClusterIP 10.110.210.164 <none> 3100/TCP 39d
loki-headless ClusterIP None <none> 3100/TCP 39d

```

```

prometheus-monitoring-kube-prometheus-prometheus-rulefiles-0 35 39d
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get configmap microservices-config -o yaml
1
apiVersion: v1
data:
  ORDER_SERVICE_URL: http://order-service:5003
  PRODUCT_SERVICE_URL: http://product-service:5002
  USER_SERVICE_URL: http://user-service:5001
kind: ConfigMap
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"ORDER_SERVICE_URL":"http://order-service:5003","PRODUCT_SERVICE_URL":"http://product-service:5002","USER_SERVICE_URL":"http://user-service:5001"},"kind":"ConfigMap","metadata":{"annotations":{},"name":"microservices-config","namespace":"default"}}
  creationTimestamp: "2026-08-30T10:32:20Z"
  name: microservices-config
  namespace: default
  resourceVersion: "13846"
  uid: 0cff2bfc-6d81-4d11-8d4e-082c42379298
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8>

```

```

PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get secrets
NAME TYPE DATA AGE
alertmanager-monitoring-kube-prometheus-alertmanager Opaque 1 39d
alertmanager-monitoring-kube-prometheus-alertmanager-cluster-tls-config Opaque 1 39d
alertmanager-monitoring-kube-prometheus-alertmanager-generated Opaque 1 39d
alertmanager-monitoring-kube-prometheus-alertmanager-tls-assets-0 Opaque 0 39d
alertmanager-monitoring-kube-prometheus-alertmanager-web-config Opaque 1 39d
loki Opaque 1 39d
loki-promtail Opaque 1 39d
microservices-secret Opaque 1 14m
mongo-secret Opaque 2 56m
monitoring-grafana Opaque 3 39d
monitoring-kube-prometheus-admission Opaque 3 39d
prometheus-monitoring-kube-prometheus-prometheus Opaque 1 39d
prometheus-monitoring-kube-prometheus-prometheus-thanos-prometheus-http-client-file Opaque 1 39d
prometheus-monitoring-kube-prometheus-prometheus-tls-assets-0 Opaque 1 39d
prometheus-monitoring-kube-prometheus-prometheus-web-config Opaque 1 39d
sh.helm.release.v1.loki.v1 helm.sh/release.v1 1 39d
sh.helm.release.v1.monitoring.v1 helm.sh/release.v1 1 39d
PS C:\Users\aaroh\OneDrive\Desktop\devtemp\Devops-Lab-L1_2023-27\23070122004_AarohiKondpalle\Project8> kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
frontend-service 1/1 1 1 7m3s
mongo-deployment 1/1 1 1 54m
mongo-express-deployment 1/1 1 1 49m
monitoring-grafana 0/1 1 0 39d
monitoring-kube-prometheus-operator 1/1 1 1 39d
monitoring-kube-state-metrics 1/1 1 1 39d
nginx 1/1 1 1 39d
order-service 1/1 1 1 8m15s
product-service 1/1 1 1 10m
social-media-app 1/1 1 1 13d

```

The application was successfully containerized into **4 microservices** and deployed on Kubernetes using Deployments and Services.

ConfigMaps and Secrets were used for configuration and secure data management.

The microservices successfully communicated with each other, and the application was verified through the frontend.